



PRODUCT NAME

Advanced Second Test Formula

Description

A professionally crafted formulation designed as an effective moisturizing and protective cream. This product is cost-effective, white in color, medium viscosity, and fragrance-free to cater for sensitive skin.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-1000 liters
Processing Time:	3 hours
Temperature:	75°C for phases A and B mixing, cool to 40°C post-emulsification
Storage Conditions:	Store in a cool, dry place, away from direct sunlight.
Equipment:	High shear mixer, heating vessels, pH meter
Certification:	Not applicable

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Purified Water	Aqua	53.0%	Solvent
2	Glycerin	Glycerin	10.0%	Humectant
3	Cetyl Alcohol	Cetyl Alcohol	5.0%	Emollient
4	Stearic Acid	Stearic Acid	8.0%	Thickener
5	Ceteareth-20	Ceteareth-20	3.0%	Emulsifier
6	Xanthan Gum	Xanthan Gum	0.5%	Stabilizer
7	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative
8	Sodium Hydroxide	Sodium Hydroxide	0.3%	pH Adjuster
9	Dimethicone	Dimethicone	5.2%	Skin Conditioning
10	Sunflower Oil	Helianthus Annuus (Sunflower) Seed Oil	4.0%	Emollient
11	Vitamin E Acetate	Tocopheryl Acetate	1.0%	Antioxidant
12	Disodium EDTA	Disodium EDTA	0.5%	Chelating Agent
Total			100%	

Manufacturing Process

Phase 1: A

1. Heat Purified Water to 75°C.
2. Disperse Xanthan Gum in heated Purified Water and mix until fully hydrated.

Phase 2: B

1. In a separate vessel, combine Stearic Acid, Cetyl Alcohol, Ceteareth-20, Dimethicone, Sunflower Oil, and heat to 75°C until melted.
2. Mix thoroughly to form a uniform blend.

Phase 3: C

1. Add Phase B (oil phase) slowly into Phase A (water phase) with continuous stirring to form an emulsion.
2. Continue to mix while cooling down the mixture to 40°C.

Phase 4: D

1. Add Glycerin, Vitamin E Acetate, and Phenoxyethanol to the emulsion.
2. Adjust pH to 7 using the Sodium Hydroxide solution.

Phase 5: E

1. Add Disodium EDTA and continue mixing until the mixture is homogeneous.
2. Allow cooling to room temperature.

Required Equipment

High shear mixer, heating vessels, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place, away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Not applicable

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-1000 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of cream to the skin as needed, and gently massage until absorbed. Suitable for use on face and body.